

Linear Algebra and Geometry

1 Lecture 1

We let $\mathbb{F}$ denote either the set of rational numbers, $\mathbb{Q}$, or the set of real numbers, $\mathbb{R}$. The elements of $\mathbb{F}$ we shall call scalars.

Definition 1.1. For a positive integer n we denote

$$[n] \stackrel{\text{def}}{=} \{1, 2, \dots, n\} \subset \mathbb{Z}.$$

Definition 1.2. An m -by- n matrix over $\mathbb{F}$ is an rectangular array of elements of $\mathbb{F}$, arranged in m rows and n columns. It can be written as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}. \quad (1.1)$$

We usually denote this array by a single capital letter (i.e., A) or by $[a_{ij}]$. We denote by $[A]_{(i)}$ and $[A]^{(j)}$ the i -th row and j -th column of a matrix A , respectively. The entry in the i -th row and j -th column of a matrix A we shall refer to as the (i, j) -th entry (or element) of A , and denote by $[A](i, j)$ or a_{ij} . If A is a matrix having m rows and n columns then A is said to have order m -by- n or is called a matrix of size m -by- n .

Definition 1.3. The main diagonal (or principal diagonal) of an m -by- n matrix A is the collection of entries $[A](i, j)$ such that $i = j$, where $i \in [m]$ and $j \in [n]$. The main diagonal is indicated in bold type in the following example

$$B = \begin{bmatrix} \mathbf{0} & -1 & -1 & 3 \\ 2 & \mathbf{-1} & 0 & 0 \\ -1 & 3 & \mathbf{3} & -1 \\ 0 & -1 & 2 & \mathbf{-2} \\ 3 & 0 & 7 & -9 \end{bmatrix}, \quad \begin{aligned} [B](1, 1) &= 0, \\ [B](2, 2) &= -1, \\ [B](3, 3) &= 3, \\ [B](4, 4) &= -2. \end{aligned}$$

Definition 1.4. The antidiagonal (or counter diagonal) of an m -by- n matrix A is the collection of all entries $[A](i, j)$ such that $i + j = n + 1$, where $i \in [m]$ and $j \in [n]$. The antidiagonal is indicated in bold type in the following example

$$C = \begin{bmatrix} 2 & 0 & -1 & -1 & \mathbf{-3} \\ -7 & 2 & -1 & \mathbf{0} & 0 \\ 5 & -1 & \mathbf{9} & 3 & -1 \\ 3 & \mathbf{0} & -1 & 2 & -1 \end{bmatrix}, \quad \begin{aligned} [C](1, 5) &= -3, \\ [C](2, 4) &= 0, \\ [C](3, 3) &= 9, \\ [C](4, 2) &= 0. \end{aligned}$$

We denote by $\text{Mat}(m, n, \mathbb{F})$ the set of all m -by- n matrices over $\mathbb{F}$ (alternative notations for this set are $\text{Mat}(m \times n, \mathbb{F})$ or $\text{Mat}_{m,n}(\mathbb{F})$ or $\text{Mat}_{m \times n}(\mathbb{F})$).

Definition 1.5. Two matrices $A = [a_{ij}] \in \text{Mat}(k, l, \mathbb{F})$ and $B = [b_{ij}] \in \text{Mat}(m, n, \mathbb{F})$ are equal, denoted $A = B$, if and only if $k = m$, $l = n$ and $a_{ij} = b_{ij}$, for all $i \in [m]$, $j \in [n]$.

Special Matrices

Definition 1.6. A square matrix of order n is an n -by- n matrix; that is, a matrix with the same number of rows as columns.

We denote by $\text{Mat}_n(\mathbb{F})$ the set of all square matrices of order n over $\mathbb{F}$ (alternative notations for this set is $M_n(\mathbb{F})$).

Definition 1.7. An m -by- n matrix $A = [a_{ij}]$ over $\mathbb{F}$ is said to have

- 1) an upper triangular form if $a_{ij} = 0$, for $i > j$, where $i \in [m]$, $j \in [n]$;
- 2) a lower triangular form if $a_{ij} = 0$, for $i < j$, where $i \in [m]$, $j \in [n]$.

For example, let asterisks, $*$, denote any elements of $\mathbb{F}$ then the left 4-by-5 matrix has an upper triangular form, and the right one has a lower triangular form:

$$\begin{bmatrix} * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \end{bmatrix}, \quad \begin{bmatrix} * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 \\ * & * & * & 0 & 0 \\ * & * & * & * & 0 \end{bmatrix}.$$

Definition 1.8. A matrix $A \in \text{Mat}_n(\mathbb{F})$ is said to be

- 1) an upper triangular matrix if $a_{ij} = 0$, for $i > j$, where $i, j \in [n]$;
- 2) a lower triangular matrix if $a_{ij} = 0$, for $i < j$, where $i, j \in [n]$;
- 3) a triangular matrix if it is an upper or a lower triangular matrix.

Definition 1.9. An m -by- n matrix $A = [a_{ij}]$ over $\mathbb{F}$ is called diagonal, denoted $\text{diag}_{m,n}(a_{11}, a_{22}, \dots, a_{pp})$, where $p = \min(m, n)$, if $a_{ij} = 0$ for $i \neq j$, where $i \in [m]$ and $j \in [n]$. That is, one may say that a diagonal matrix is a matrix in which the entries outside the main diagonal are all zero. Also, one may say that a diagonal matrix is a matrix having both upper and lower triangular form.

Note that the entries a_{ii} , for $i \in [p]$, are unrestricted. If $A = [a_{ij}]$ is a diagonal matrix and $m = n$ then we denote it by $\text{diag}_n(a_{11}, a_{22}, \dots, a_{pp})$. For example, the first three of the following matrices are diagonal, and the fourth one is not

$$\text{diag}_{2,3}(-2, 1) = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \text{diag}_2(3, 7) = \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix}, \quad \text{diag}_{4,3}(0, 0, 0) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Definition 1.10. If $A = \text{diag}_{m,n}(a_{11}, a_{22}, \dots, a_{pp})$ and $a_{ii} = \lambda$, for $i \in [p]$, then the diagonal matrix A is called a scalar matrix.

Definition 1.11. The identity matrix of order n over $\mathbb{F}$, denoted I_n , is defined as:

$$I_n \stackrel{\text{def}}{=} \text{diag}_n(1, 1, \dots, 1).$$

That is, I_n a square matrix of order n with $a_{ii} = 1$, for $i \in [n]$, and zeros elsewhere. For example

$$I_1 = [1], \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Definition 1.12. The m -by- n zero matrix (or null matrix) over $\mathbb{F}$, denoted $0_{m,n}$, is a matrix all of whose entries are equal to $0 \in \mathbb{F}$. If $m = n$, then we use the notation 0_n instead of the notation $0_{n,n}$. For example:

$$0_{2,3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad 0_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Definition 1.13. A matrix having only one column is called a column vector and a matrix with only one row is called a row vector. Usually our row and column vectors will be represented by bold letters. For example

$$\mathbf{a} = [a_1 \quad a_2 \quad \dots \quad a_n] \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Operations with matrices

1. Scalar multiplication

Definition 1.14. The product of a scalar $\lambda \in \mathbb{F}$ and a matrix $A = [a_{ij}] \in \text{Mat}(m, n, \mathbb{F})$ is the m -by- n matrix over $\mathbb{F}$, denoted λA , whose (i, j) -th entry, $[\lambda A](i, j)$, is

$$\begin{aligned} [\lambda A](i, j) &\stackrel{\text{def}}{=} \lambda \cdot [A](i, j) \\ &= \lambda a_{ij}. \end{aligned} \tag{1.2}$$

That is, λA is the matrix whose entries are λ times each of the entries of A .

This operation is called scalar multiplication of matrices (more precisely, left scalar multiplication of matrices; the right one is defined as $[A\lambda](i, j) \stackrel{\text{def}}{=} a_{ij}\lambda$; it should be clear that $\lambda A = A\lambda$). For example:

$$2 \begin{bmatrix} 0 & -1 \\ 2 & -2 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 2 \cdot 0 & 2 \cdot (-1) \\ 2 \cdot 2 & 2 \cdot (-2) \\ 2 \cdot (-1) & 2 \cdot 7 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 4 & -4 \\ -2 & 14 \end{bmatrix}.$$

2. Matrix addition

Definition 1.15. The sum of two matrices $A = [a_{ij}], B = [b_{ij}] \in \text{Mat}(m, n, \mathbb{F})$ is the m -by- n matrix over $\mathbb{F}$, denoted $A + B$, whose (i, j) -th entry, $[A + B](i, j)$, is

$$\begin{aligned} [A + B](i, j) &\stackrel{\text{def}}{=} [A](i, j) + [B](i, j) \\ &= a_{ij} + b_{ij}. \end{aligned} \tag{1.3}$$

That is, $A + B$ is the matrix whose entries are the sums of the corresponding entries of A and B .

This operation is called matrix addition (more precisely, component-wise matrix addition). Note that two matrices over $\mathbb{F}$ can be added if and only if they have an equal number of rows and columns. For example:

$$\begin{bmatrix} 0 & -1 \\ 2 & -2 \\ -1 & 7 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0+2 & -1+1 \\ 2-1 & -2+3 \\ -1-1 & 7+0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ -2 & 7 \end{bmatrix}.$$

The following statement is an exercise.

Statement 1.1. *The following rules are satisfied by the operations of matrix addition and scalar multiplication for any $A, B, C \in \text{Mat}(m, n, \mathbb{F})$ and $0, 1, -1, \lambda_1, \lambda_2 \in \mathbb{F}$:*

- 1) $A + B = B + A;$ *(matrix addition is commutative)*
- 2) $(A + B) + C = A + (B + C);$ *(matrix addition is associative)*
- 3) $0_{m,n} + A = A = A + 0_{m,n};$ *(zero matrix is the neutral element)*
- 4) $A + (-1)A = 0_{m,n};$ *(there exists the additive inverse)*
- 5) $(\lambda_1 + \lambda_2)A = \lambda_1 A + \lambda_2 A;$
- 6) $\lambda_1(A + B) = \lambda_1 A + \lambda_1 B;$
- 7) $(\lambda_1 \lambda_2)A = \lambda_1(\lambda_2 A);$
- 8) $1 \cdot A = A$
- 9) $0 \cdot A = 0_{m,n}.$

3. Matrix multiplication

Definition 1.16. The product of two matrices $A = [a_{ij}] \in \text{Mat}(m, n, \mathbb{F})$ and $B = [b_{ij}] \in \text{Mat}(n, p, \mathbb{F})$ is the m -by- p matrix over $\mathbb{F}$, denoted AB , whose (i, j) -th entry, $[AB](i, j)$, is

$$\begin{aligned} [AB](i, j) &\stackrel{\text{def}}{=} \sum_{k \in [n]} [A](i, k) \cdot [B](k, j) \\ &= \sum_{k \in [n]} a_{ik} \cdot b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}. \end{aligned} \quad (1.4)$$

That is, (i, j) -th entry of AB is obtained by taking each entry of row i of A and multiplying it by the corresponding entry of column j of B , then adding these n products together.

This operation is called matrix multiplication (or matrix product). Note that the product of two matrices is defined if and only if the number of columns of the left matrix is the same as the number of rows of the right matrix. For example:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 5 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 0 \cdot 0 + 2 \cdot 1 & 1 \cdot (-1) + 0 \cdot 5 + 2 \cdot 1 \\ 0 \cdot 2 + (-1) \cdot 0 + 3 \cdot 1 & 0 \cdot (-1) + (-1) \cdot 5 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 3 & -2 \end{bmatrix}.$$

It is an important consequence of Formula (1.4) that even if the products AB and BA are both defined (i.e., $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, m, \mathbb{F})$) it need not be true that $AB = BA$; in other words, matrix multiplication is *not commutative*. Indeed:

$$\begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

Corollary 1.1. *Let $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, p, \mathbb{F})$. Then, for any $i \in [m]$ and $j \in [p]$:*

$$[AB]_{(i)} = \sum_{k \in [n]} [A](i, k) \cdot [B]_{(k)}; \quad (1.5)$$

$$= [A]_{(i)} B; \quad (1.6)$$

$$[AB]^{(j)} = \sum_{k \in [n]} [A]^{(k)} \cdot [B](k, j); \quad (1.7)$$

$$= A[B]^{(j)}. \quad (1.8)$$

Statement 1.2. *The following rules are satisfied by the operations of matrix addition, scalar multiplication and matrix multiplication:*

1) *matrix multiplication is distributive with respect to matrix addition, that is,*

$$\begin{aligned} A(B + C) &= AB + AC && \text{for any } A \in \text{Mat}(m, n, \mathbb{F}) \text{ and } B, C \in \text{Mat}(n, p, \mathbb{F}) \text{ (left distributivity);} \\ (B + C)A &= BA + CA && \text{for any } A \in \text{Mat}(n, p, \mathbb{F}) \text{ and } B, C \in \text{Mat}(m, n, \mathbb{F}) \text{ (right distributivity);} \end{aligned}$$

2) *matrix multiplication is associative, that is,*

$$(AB)C = A(BC) \quad \text{for any } A \in \text{Mat}(m, n, \mathbb{F}), B \in \text{Mat}(n, p, \mathbb{F}) \text{ and } C \in \text{Mat}(p, q, \mathbb{F});$$

3) *matrix multiplication is compatible with scalars, that is,*

$$(\lambda_1 A)(\lambda_2 B) = (\lambda_1 \lambda_2)(AB) \quad \text{for any } A \in \text{Mat}(m, n, \mathbb{F}), B \in \text{Mat}(n, p, \mathbb{F}) \text{ and } \lambda_1, \lambda_2 \in \mathbb{F};$$

4) I_n *is the multiplicative identity of* $M_n(\mathbb{F})$, *that is,*

$$I_n A = A = A I_n \quad \text{for any } A \in \text{Mat}_n(\mathbb{F}).$$

Proof. 1) To prove left distributivity, by Definition 1.5, it suffices to show that the corresponding entries of the matrices $A(B + C)$ and $AB + AC$ are the same. Indeed, for any $i \in [m]$ and $j \in [p]$ we have (the number above the equal sign denotes the equality we used):

$$\begin{aligned} [A(B + C)](i, j) &\stackrel{(1.4)}{=} \sum_{k \in [n]} [A](i, k) \cdot [B + C](k, j) \\ &\stackrel{(1.3)}{=} \sum_{k \in [n]} [A](i, k) \cdot ([B](k, j) + [C](k, j)) \\ &= \sum_{k \in [n]} [A](i, k) \cdot [B](k, j) + \sum_{k \in [n]} [A](i, k) \cdot [C](k, j) \\ &\stackrel{(1.4)}{=} [AB](i, j) + [AC](i, j) \\ &\stackrel{(1.3)}{=} [AB + AC](i, j). \end{aligned}$$

The equality $(B + C)A = BA + CA$ can be proven similarly.

2) It suffices to prove that $[(AB)C](i, j) = [A(BC)](i, j)$, where $i \in [m]$ and $j \in [q]$. Indeed, for any

$i \in [m]$ and $j \in [q]$ we have:

$$\begin{aligned}
[(AB)C](i, j) &\stackrel{(1.4)}{=} \sum_{k \in [p]} \left([AB](i, k) \cdot [C](k, j) \right) \\
&\stackrel{(1.4)}{=} \sum_{k \in [p]} \left(\sum_{l \in [n]} \left([A](i, l) \cdot [B](l, k) \right) \cdot [C](k, j) \right) \\
&= \sum_{k \in [p]} \left(\sum_{l \in [n]} \left([A](i, l) \cdot [B](l, k) \cdot [C](k, j) \right) \right) \\
&= \sum_{l \in [n]} \left(\sum_{k \in [p]} \left([A](i, l) \cdot [B](l, k) \cdot [C](k, j) \right) \right) \\
&= \sum_{l \in [n]} \left([A](i, l) \cdot \sum_{k \in [p]} \left([B](l, k) \cdot [C](k, j) \right) \right) \\
&\stackrel{(1.4)}{=} \sum_{l \in [n]} \left([A](i, l) \cdot [BC](l, j) \right) \\
&\stackrel{(1.4)}{=} [A(BC)](i, j).
\end{aligned}$$

3) and 4) The proof of these statements is left as an exercise. ■